

The Fine-Structure Constant as a Charge Count on One Geometry

Standalone companion to the flagship — DRAFT, Keeper, 2026-07-18. The single most checkable BST result: $\alpha^{-1} = 137.0365$ with no free normalization. A referee can verify it in an afternoon.

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Abstract

We give a parameter-free geometric account of the fine-structure constant. On the bounded symmetric domain $D_{IV}^5 = SO(5,2)/[SO(5) \times SO(2)]$, electric charge is the integer weight of the $SO(2)$ circle, and the number of independent charge modes is a combinatorial capacity — a democratic dimension count, independent of any norm: $N_{\max} = N_c^3 \cdot n_C + \text{rank} = 137$. The factor 4π that appears in α 's definition is identified as the Coulomb solid angle of the geometric descent $SO(5,2) \rightarrow SO(3,1)$ (Gauss's law in three spatial dimensions), so α is a pure number for a geometric reason, not by convention. A single curvature correction, $n_C/N_{\max}$, refines the integer to

$$\alpha^{-1} = N_{\max} + \frac{n_C}{N_{\max}} = 137 + \frac{5}{137} = 137.036496,$$

against the measured 137.035999 — a deviation of 0.0004%. The three integers ($N_c = 3$, $n_C = 5$, $\text{rank} = 2$) are structural invariants of the domain, fixed before any comparison to 137; the same curvature correction independently closes the Higgs mass, a two-target check that rules out post-hoc tuning. A measured α^{-1} inconsistent with $137 + 5/137$ at this precision falsifies the account.

1. The result

$$\alpha^{-1} = N_{\max} + n_C/N_{\max} = 137 + 5/137 = 137.036496 \quad (\text{obs } 137.035999, 0.0004\%)$$

Three ingredients, each geometric: an integer capacity (Section 2), the reason α carries a 4π (Section 3), and one curvature correction (Section 4). No fitted normalization enters.

2. The integer 137 as a charge capacity

Electric charge in BST is the integer weight of the intrinsic $SO(2)$ (the “charge circle,” the complex structure of the domain; T2470). The coupling is set not by the *norm* of a charged state but by how many independent charge modes the domain supports — a democratic count, combinatorial and norm-independent (this is why the 35× spread in charge-block norms is irrelevant to α ; K678). That count is

$$N_{\max} = N_c^3 \cdot n_C + \text{rank} = 27 \cdot 5 + 2 = 137.$$

- $N_c^3 = 27$ — the color modes cubed (the Albert/exceptional 27; the color sector's dimension count). - $n_C = 5$ — over the five complex dimensions of the domain. - $\text{rank} = 2$ — the two Shilov “ruler” directions.

Tier: the integer $137 = N_{\max}$ is a structural count [derived], and it is the same $N_{\max}$ that appears across the corpus (it is not introduced to fit α). Whether *this specific* decomposition ($27 \cdot 5 + 2$) is the unique combinatorial reading is the one presentation-level open point (Cal referee).

3. Why α carries a 4π — the Coulomb solid angle

α is conventionally written with a 4π (rationalized units: $\alpha = e^2/4\pi\epsilon_0\hbar c$). BST reads this 4π geometrically: it is the solid angle of 3-space, produced by the descent $SO(5,2) \rightarrow SO(4,2) \rightarrow SO(3,1)$ that yields 3+1 spacetime (Casey #14). Charge measured through the descent is measured over the full Coulomb solid angle 4π — Gauss's law (Casey's reading, K699). So the 4π is not a unit convention we happen to divide by; it is the geometric fact that the emergent space is three-dimensional. This is why α is a pure number: capacity (a count) over solid angle (a geometric constant). [derived-framework]

4. The curvature correction $n_C/N_{\max}$

The integer capacity is the flat count; the domain is curved, and the leading curvature correction to the count is

$$\frac{n_C}{N_{\max}} = \frac{5}{137} = 0.036496,$$

the norm-weighted (Bergman/heat-kernel) curvature contribution over the same capacity (T2133; K701). It shifts $137 \rightarrow 137.0365$, matching observation to 0.0004%. [derived; the correction's coefficient is n_C , not fitted]

5. Target-innocence (why this is not numerology)

Two independent guards: 1. The integers are structural. N_c , n_C , rank are the invariants of D_{IV}^5 (the rank-2 root data; the domain is fixed by (3 colors, 3 generations, dim 5)). They are pinned *before* any comparison to 137 — the value 137 is target-innocent in the sense of the derived-vs-fit discipline. 2. The two-target check. The *same* curvature correction that closes α also closes the Higgs mass (E1/K701). A tuned correction fits one target; a geometric one closes two with the same coefficient. This is the decisive check against post-hoc tuning. And a retired trap, stated for honesty: the old Shannon/Wyler numerology route to 137 is not resurrected here — that route relocates under the norm-weighted measure and is retired (K676). The present account is the capacity-count, not the Wyler integral.

6. What is not claimed

- The $4\pi = \text{Coulomb-solid-angle}$ identification is *framework-level* (it explains why α is pure and dimensionless), not a separate numerical derivation.
- The uniqueness of the $27 \cdot 5 + 2$ decomposition of 137 is a presentation point for the referee.
- α is a *coupling*; at other scales it runs (standard QED). The 137 is the low-energy value; the geometric account is of that value, not of the running (which is standard).

7. Falsifiability and conclusion

The claim is sharp: $\alpha^{-1} = 137 + 5/137 = 137.0365$, from a charge count ($137 = N_c^3 \cdot n_C + \text{rank}$), a solid angle (4π , three-dimensionality), and one curvature correction ($n_C/N_{\max}$) — no free normalization. A measured α^{-1} that departs from $137 + 5/137$ at the 0.0004% level, or a demonstration that the integers are fit rather than structural, falsifies it. The fine-structure constant, on this account, is not a mystery number: it is *how many ways charge fits on the geometry, seen through the solid angle of the space the geometry makes*.

Draft status: complete as a short standalone; the single companion to the flagship (BST_FLAGSHIP...) and the Five-Absence falsifier paper. Needs: Cal referee on the $27 \cdot 5 + 2$ uniqueness (Section 2) and the curvature-coefficient (Section 4); the explicit two-target Higgs computation inlined from E1. Number verified: $137 + 5/137 = 137.036496$ vs obs 137.035999, 0.0004%.

— Keeper, 2026-07-18. $\alpha^{-1} = N_{\max} + n_C/N_{\max} = 137 + 5/137 = 137.0365$ (0.0004%). Capacity $137 = N_c^3 \cdot n_C + \text{rank}$ (charge count, norm-independent); $4\pi = \text{Coulomb solid angle of the 3D descent}$ (α pure for a geometric reason); curvature correction $n_C/N_{\max}$; target-innocent (structural integers + two-target Higgs check). See [[BST_FLAGSHIP_The_Standard_Model_as_Representation_Theory_of_D_IV5_DRAFT_2026-07-18]], [[Keeper_Strengthening_Roadmap_effort_ranked_2026-07-18]].